MARINE AI INTELLIGENCE: TECHNOLOGY STACK ARCHITECTURE REPORT

Autonomous Acoustic Hydrographic Intelligence & Debris Classification Engine

Smart India Hackathon (SIH 2026) · Ministry of Earth Sciences (MoES)

Document Ref: MoES/SIH-2026/TECHSTACK-SPEC-v1.0	**Scope:** Frontend + Backend + Simulation + Algorithms
Date: September 6, 2026	**Status:** Production Build Verified (Zero Errors)

1. Technology Stack Breakdown by Layer

Tier	Technology Stack	Purpose & Operational Scope
Frontend Framework	React 18 / 19, TypeScript 6.0	Component architecture with strict static typing for sonar geometry and telemetry.
Bundler & Tooling	Vite 8.2.2 (Rollup)	Fast HMR and optimized production build (built in 1.05s).
Styling & Theme	Tailwind CSS v4, PostCSS	Utility styling with CSS variables for seamless Dark and Light theme switching.
Interactive Canvas	HTML5 2D Canvas API	2048x1024 waterfall rendering, 800% zoom, minimap, and calibrated metric ruler.
XAI Vector Charts	Custom Mathematical SVG	Animated 5-axis Spider Radar, 2D FFT spectrum graph, and 2.5D shadow schematic.
Backend REST API	Spring Boot 3.x, Java 21 LTS	Endpoints (/api/analyze, /detections, /statistics) with multi-threaded execution.
Computer Vision	OpenCV 4.x (Java Bindings)	Native C++ acceleration for CLAHE contrast equalization and morphological filtering.
Memory Management	MatScope (AutoCloseable)	Deterministic native C++ memory release preventing memory leaks during surveys.
Physics Simulator	Python 3.10+, NumPy, SciPy	Side-scan forward model: grazing angle, ray-traced shadow, Rayleigh speckle, anti-aliasing.
5 Signatures Engine	Python & Java Implementations	Periodicity (FFT mesh pitch), Linearity (SVD), Circularity, Angularity, Irregularity.
Datasets & Formats	YOLO, JSON Manifests, PNG	2048x1024 grayscale sonar swaths (sim_00000 to sim_00002) at 2cm/px resolution.

2. Pipeline Latency & Computational Profile

Micro-profiled end-to-end execution time: **130.7 ms per 2048x1024 frame (~7.65 FPS)**:

- **Preprocessing (CLAHE & Speckle):** 13.5 ms | **Proposals (Adaptive Thresh):** 26.2 ms
- **5 Geometric Signatures:** 48.9 ms | **Shadow Height Profiling:** 16.7 ms
- **Texture Baseline Variance:** 20.3 ms | **Risk Assessment & Gear Inference:** 5.1 ms